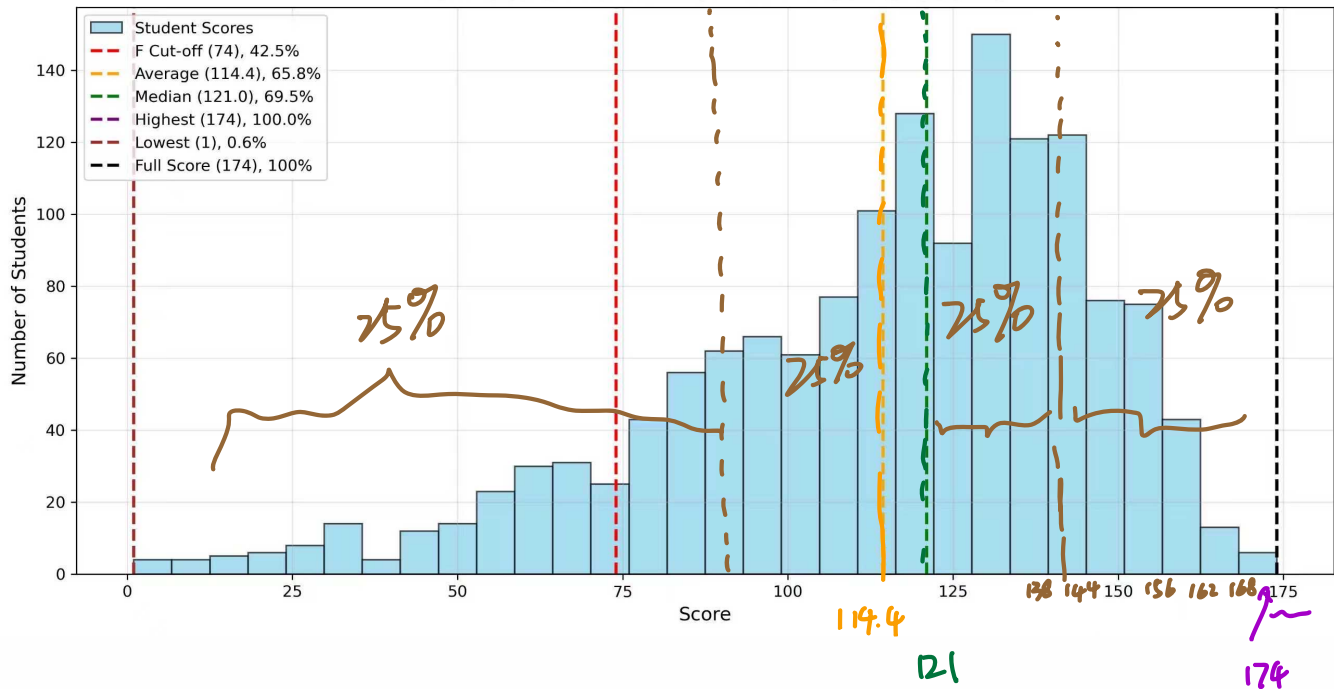
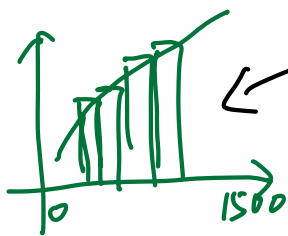


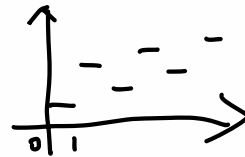
Exam Score Distribution - All 1400+ Students



$$\text{Average} = \frac{1}{1500} \sum_{i=1}^{1500} \text{Score } (i)$$



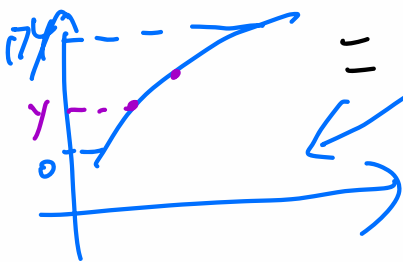
$$= \frac{1}{1500} \int_0^{1500} S(x) dx$$



$$= \sum_{j=0}^{174} j \cdot \# \{ \text{students with score } j \} \cdot \frac{1}{1500}$$

$$= \sum_{j=0}^{174} j \cdot g(j) \cdot 1 \quad (\text{where } g(j) \triangleq \frac{\# \{ \text{students with } j \}}{1500})$$

represents a frequency func)

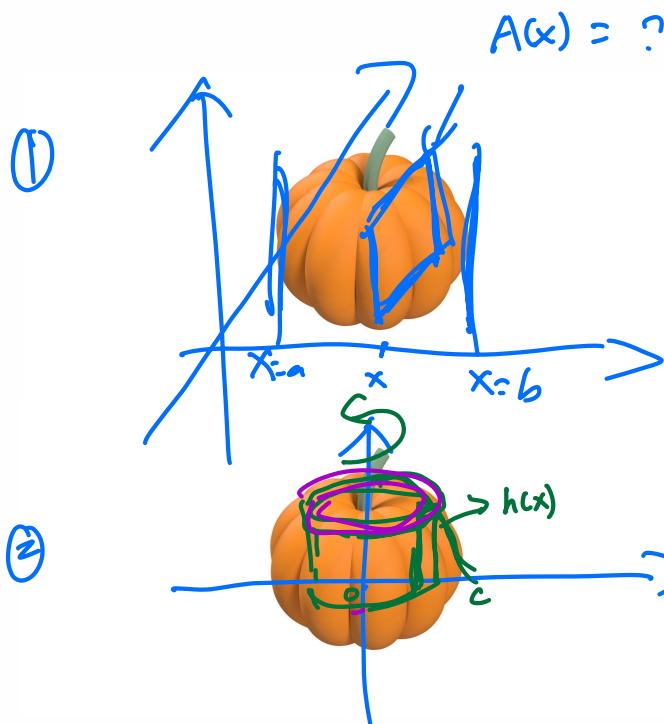
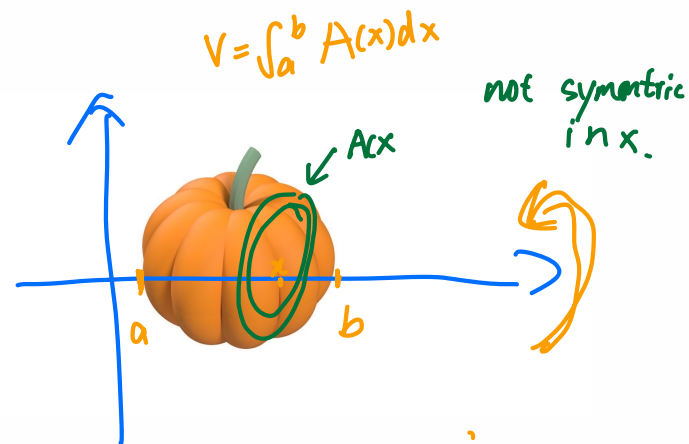
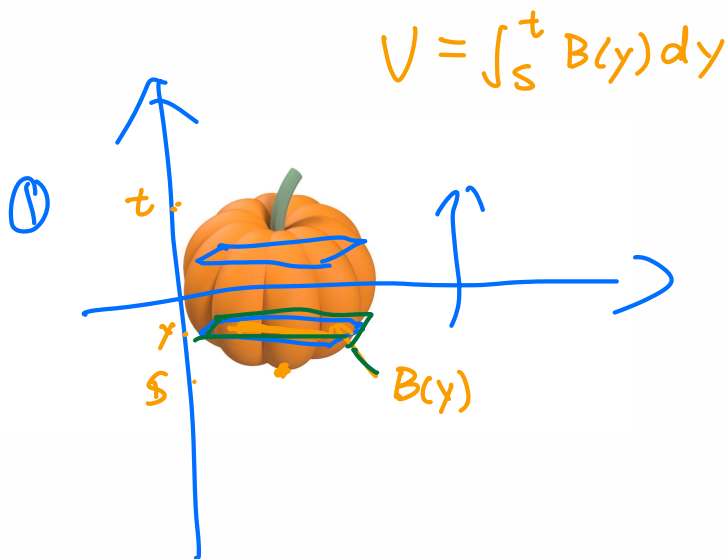


$$= \int_0^{174} y \cdot g(y) dy : \text{STA 1001}$$

MAT 2050

Lec 8-1, Oct 27, Mon

- Cylindrical shells (6.2)
- Arc length (6.3)
- Areas of surfaces of revolution (6.4)
- Arc length differential (6.3)
- Work (6.5)

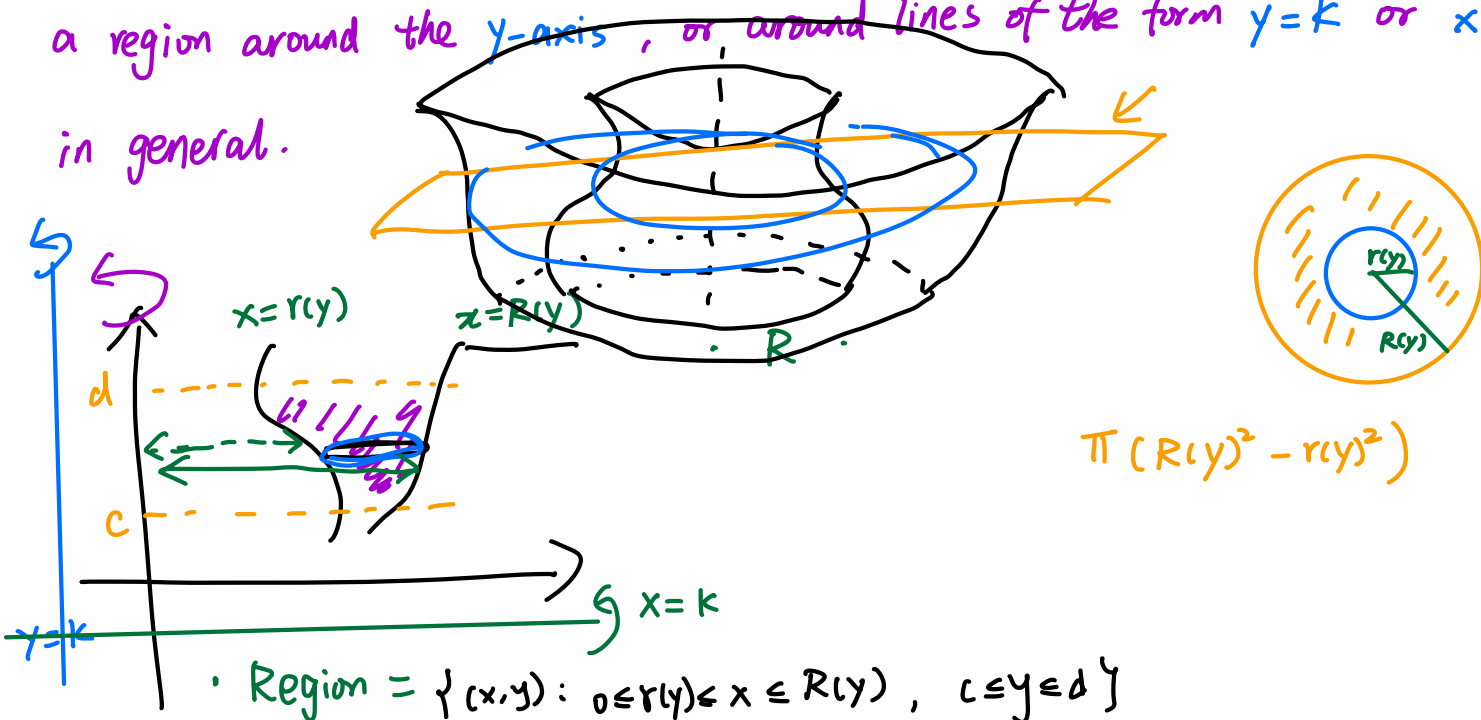


← What is the volume?

$$V = \int_a^b A(x) dx$$

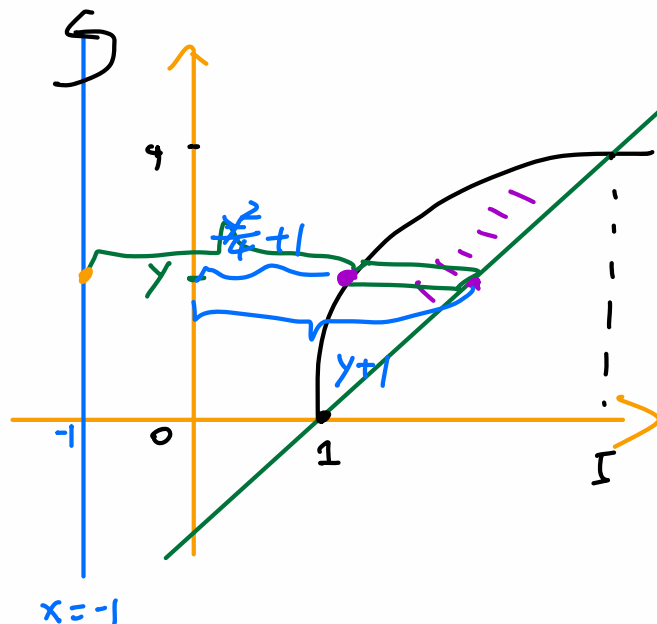
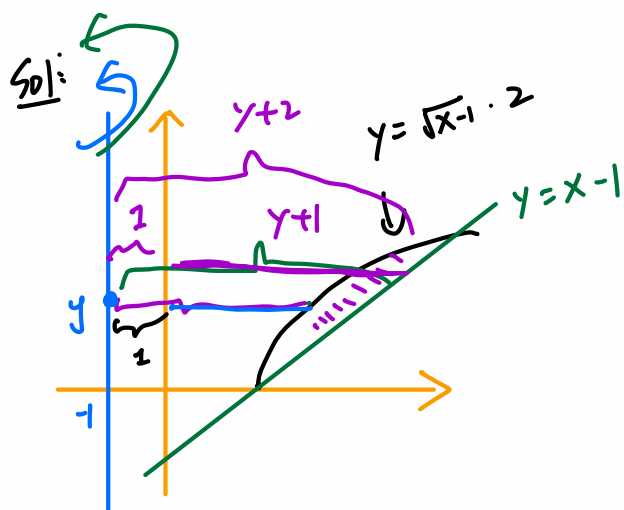
$$V = \int_0^c \underline{2\pi x} \cdot \underline{h(x)} dx. \text{ (symmetric)}$$

The ideas above can be applied similarly to a solid obtained by revolving a region around the y -axis, or around lines of the form $y=k$ or $x=k$ in general.



$$\Rightarrow V = \int_c^d A(y) dy = \int_c^d \pi (R(y)^2 - r(y)^2) dy.$$

e.g. Find the volume of the solid obtained by rotating the region bounded by $y = \sqrt{x-1} \cdot 2$ and $y = x-1$ about $x = -1$.



Step 1 : intersection:

$$\text{Let } y = x-1 = 2\sqrt{x-1} \Rightarrow x = 1 \text{ and } x = 5 \Rightarrow y = 0 \text{ and } y = 4.$$

Step 2. write down the region:

$$\text{Region} = \{ (x, y) : 1 \leq x \leq 5, x-1 \leq y \leq 2\sqrt{x-1} \}.$$

Step 3. Equivalent region:

$$\text{Region} = \{ (x, y) : 0 \leq y \leq 4, \frac{y^2}{4} + 1 \leq x \leq y+1 \}.$$

Step 4. Cross-sections and volume

$$V = \int_0^4 \pi [R(y)^2 - r(y)^2] dy$$

dy

$$= \int_0^4 \pi [(y+1 - (-1))^2 - (\frac{y^2}{4} + 1 - (-1))^2] dy$$

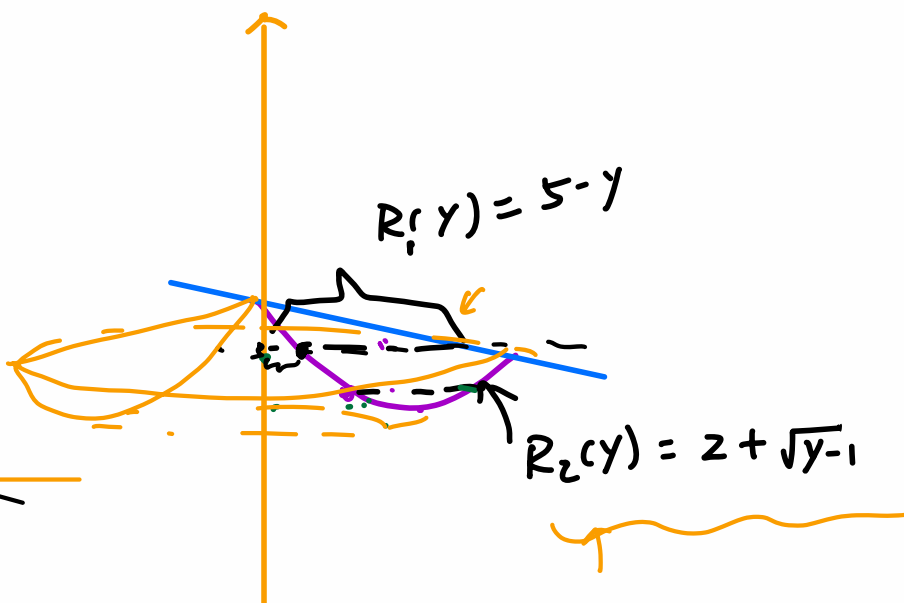
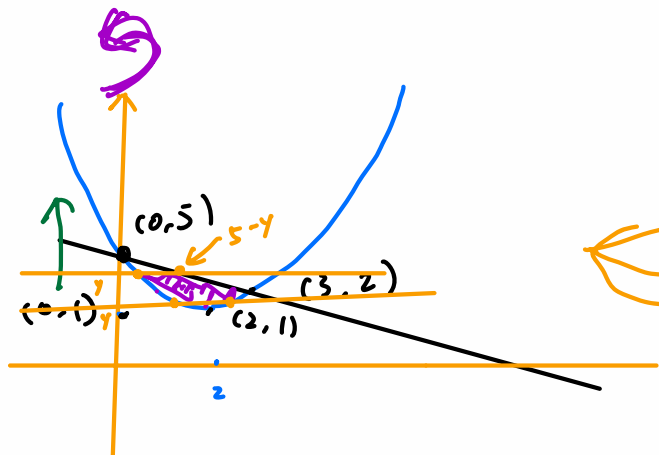
$$= \int_0^4 \pi (4y - \frac{1}{16} y^4) dy$$

$$= \frac{96}{5} \pi.$$

□.

e.g. A region is bounded by $y = (x-2)^2 + 1$ and $y = 5-x$.

Find the volume of the solid generated by revolving the region around the y-axis.



Sol': Step 1. intersection!

$$\text{Let } (x-2)^2 + 1 = 5 - x \Rightarrow x^2 - 3x = 0 \Rightarrow \underline{x=0} \text{ or } \underline{x=3}.$$

Step 2: write down the region.

$$\text{Region} = \{ (x, y) : (x-2)^2 + 1 \leq y \leq 5-x, 0 \leq x \leq 3 \}$$

$$= \left\{ (x, y) : \begin{array}{l} \begin{array}{c} (2-\sqrt{y-1}) \\ r_2(y) \end{array} \leq x \leq \begin{array}{c} (2+\sqrt{y-1}) \\ R_2(y) \end{array}, 1 \leq y \leq 2 \\ \text{and} \\ \begin{array}{c} (2-\sqrt{y-1}) \\ r_1(y) \end{array} \leq x \leq \begin{array}{c} (5-y) \\ R_1(y) \end{array}, 2 \leq y \leq 5 \end{array} \right\}.$$

Step 3: cross sections and volume.

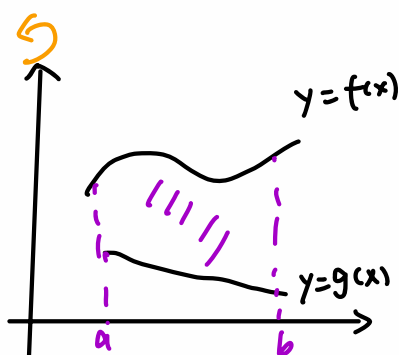
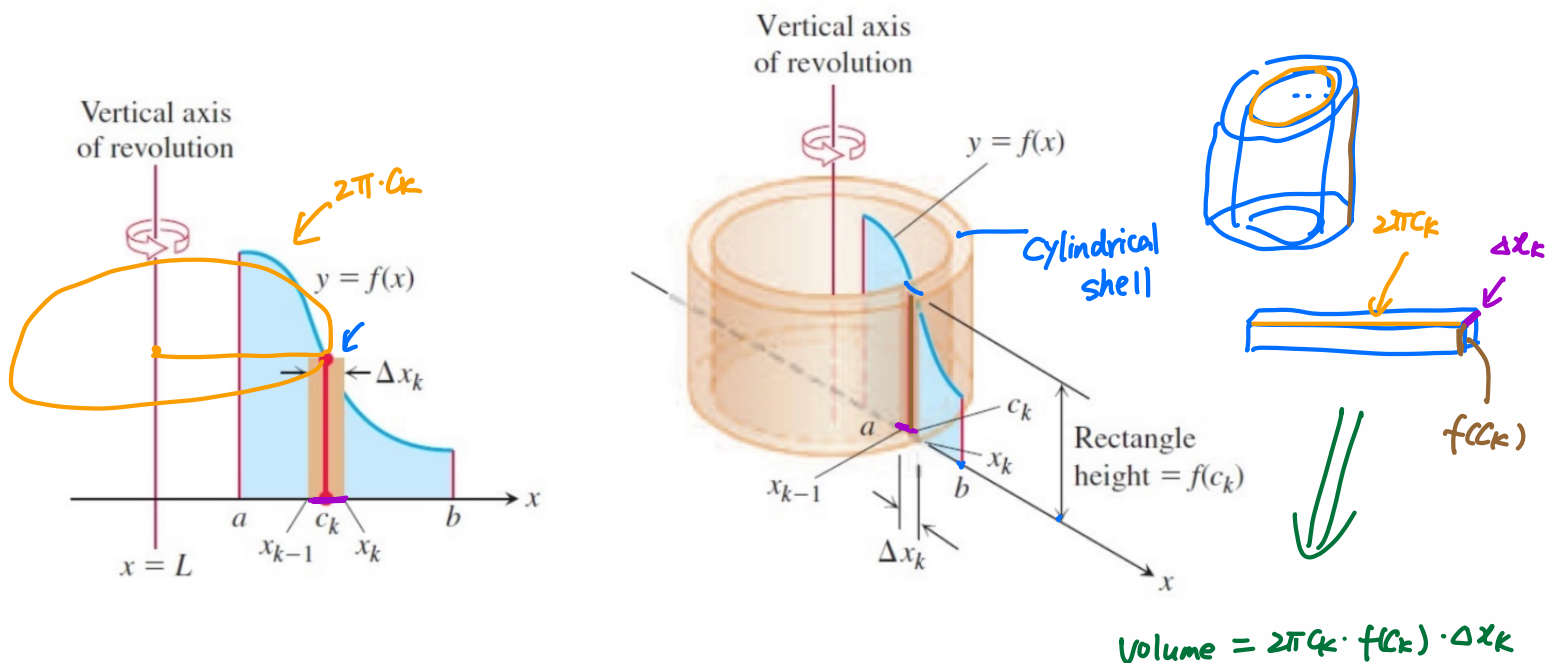
$$\begin{aligned} \therefore V &= \int_1^2 \pi (R_2^2(y) - r_2^2(y)) dy + \int_2^5 \pi (R_1^2(y) - r_1^2(y)) dy \\ &= \int_1^2 \pi ((2+\sqrt{y-1})^2 - (2-\sqrt{y-1})^2) dy + \int_2^5 \pi ((5-y)^2 - (2-\sqrt{y-1})^2) dy \\ &= \pi \left[\int_1^2 (4 + y - 1 + 4\sqrt{y-1}) - (4 - 4\sqrt{y-1} + y - 1) dy \right. \\ &\quad \left. + \int_2^5 (25 + y^2 - 10y - (4 - 4\sqrt{y-1} + y - 1)) dy \right] \\ &= \text{ugly computation} \dots \\ &= \frac{27}{2} \pi. \end{aligned}$$

□.

Volumes Using Cylindrical Shells (Optional)

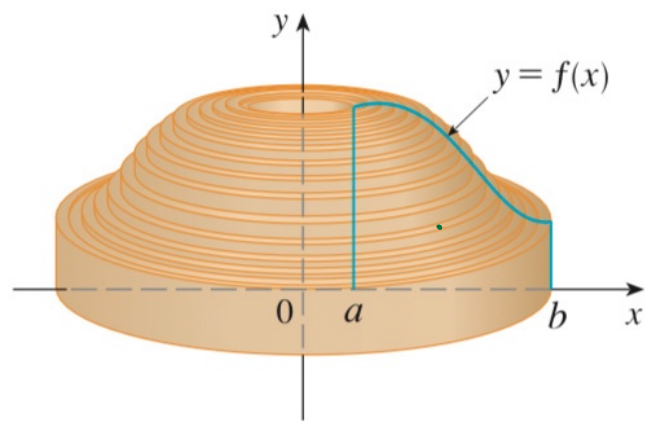
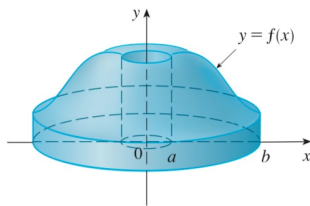
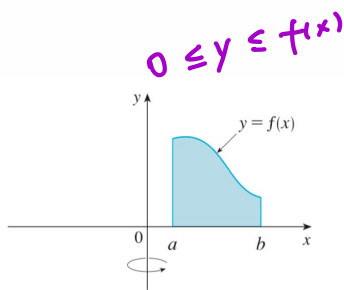
Another way to compute the volume of a solid generated by rotations around a coordinate axis is to use cylindrical shells.

Consider revolving the following region in blue about the y -axis to generate a solid. Its volume can be computed by adding the volumes of all the "cylindrical shells", one of which is displayed in orange in the figure.



- Let $h(x) = f(x) - g(x)$.
- Take $h(c_k)$ as the height, where $c_k \in [x_{k-1}, x_k]$
- Riemann sum for volume = $\sum_{k=1}^n 2\pi c_k \cdot h(c_k) \cdot \Delta x_k$
 $= \sum_{k=1}^n \text{volume of a cylindrical shell}$
- Exact volume = the limit of the Riemann sums

$$\int_a^b 2\pi x h(x) dx$$

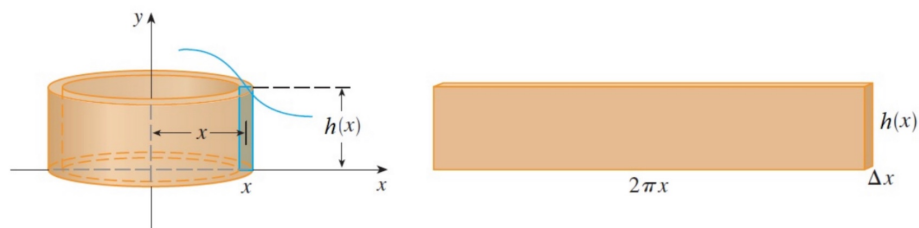


In general, given the solid S generated by revolving the region

$$\{(x, y) : g(x) \leq y \leq f(x), a \leq x \leq b\}$$

about the y -axis, let $h(x) := f(x) - g(x)$ be the height of the region at x . Then the volume V of S can be computed by

$$V = \int_a^b 2\pi x h(x) dx.$$

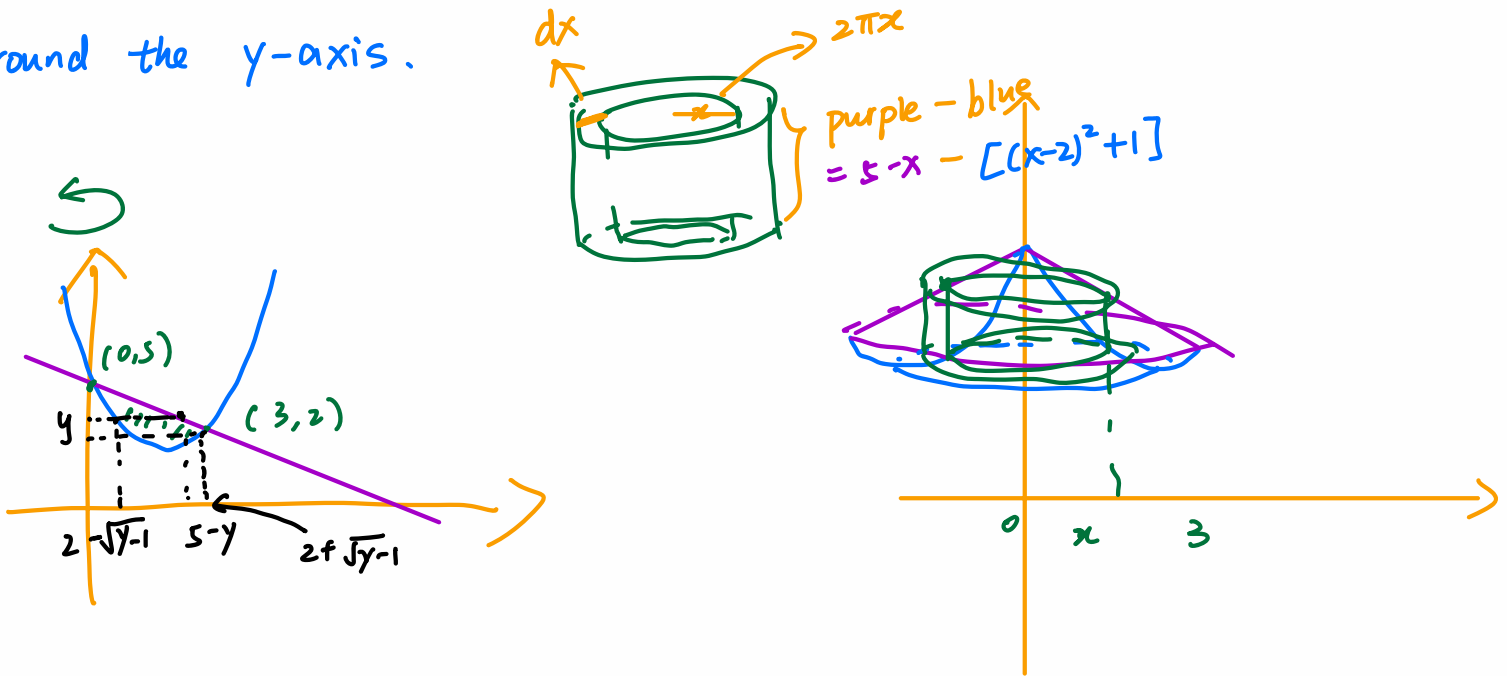


Circular ring
(cylindrical shell)

cuboid
($= 2\pi x \cdot \Delta x \cdot h(x)$)

e.g. A region is bounded by $y = (x-2)^2 + 1$ and $y = 5-x$.

Find the volume of the solid generated by revolving the region around the y -axis.



$$\begin{aligned}
 \underline{\text{Sol}}: V &= \int_0^3 2\pi x \cdot h(x) dx \\
 &= \int_0^3 2\pi x (5-x - ((x-2)^2 + 1)) dx \\
 &= 2\pi \int_0^3 x (5-x - x^2 + 4x - 5) dx \\
 &= 2\pi \int_0^3 x (+3x - x^2) dx \\
 &= 2\pi \int_0^3 -x^3 + 3x^2 dx \\
 &= 2\pi \left(-\frac{1}{4}x^4 + x^3 \right) \Big|_0^3 \\
 &= 2\pi \left(-\frac{81}{4} + 27 \right) \\
 &= \frac{27}{2} \pi .
 \end{aligned}$$

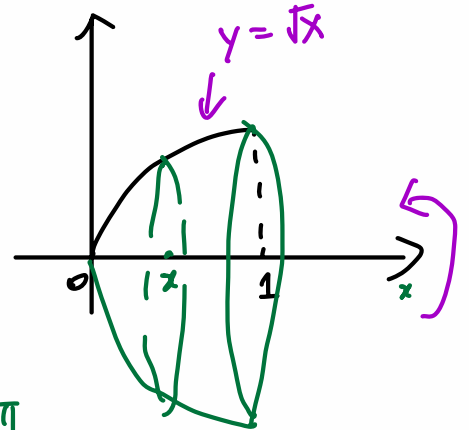
□.

[easier than the method of revolution in this example].

e.g. Find the volume of the solid generated by revolving the region between $y = \sqrt{x}$ and $y = 0$ from 0 to 1, around x -axis. Use two different methods.

Ans : $\pi/2$

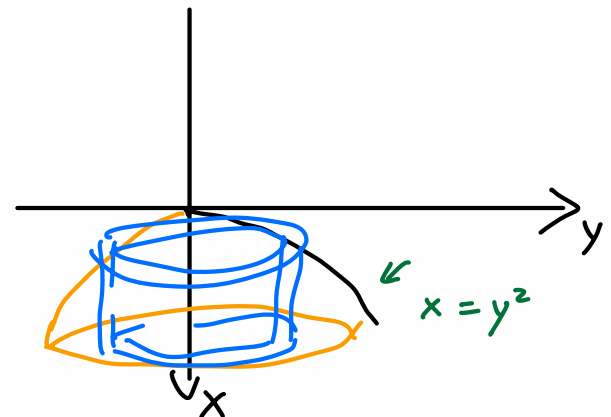
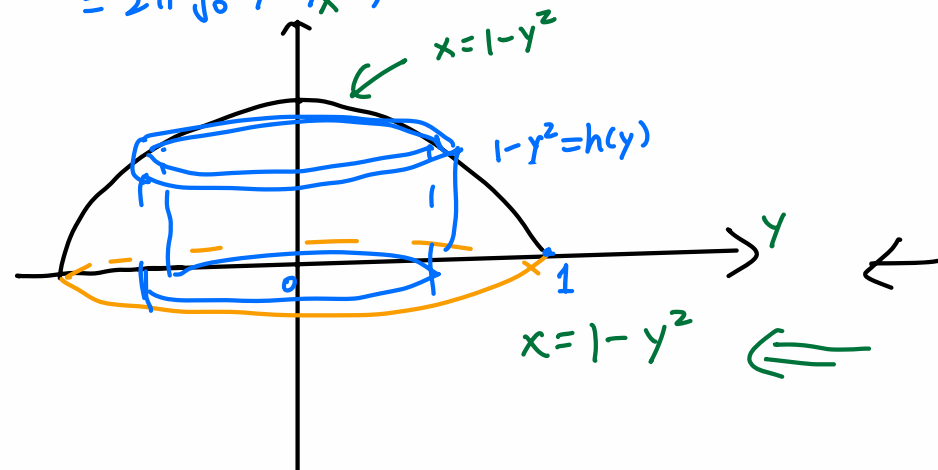
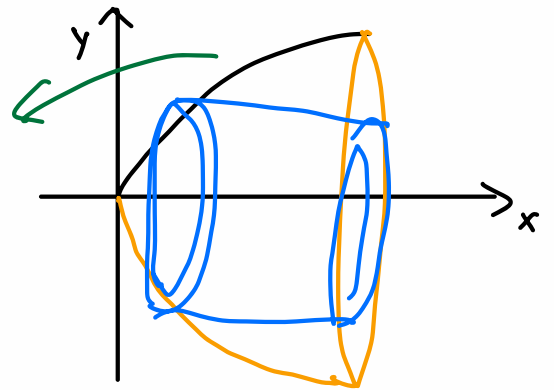
Approach I. $V = \int_0^1 A(x) dx$
 $= \int_0^1 \pi R(x)^2 dx$
 $= \int_0^1 \pi (\sqrt{x})^2 dx = \int_0^1 \pi x dx = \frac{\pi}{2}.$



Approach II.

$V = \int_0^1 2\pi y h(y) dy$
 $= \int_0^1 2\pi y (1-y^2) dy$
 $= 2\pi \int_0^1 y - y^3 dy = 2\pi (\frac{1}{2}y^2 - \frac{1}{4}y^4) \Big|_0^1 = \frac{\pi}{2}.$

$y = \sqrt{x}$
 $\Rightarrow y^2 = x$



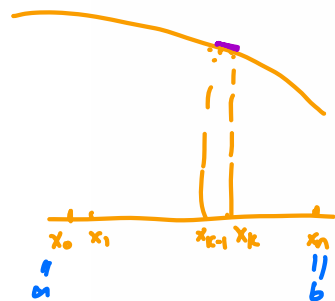
Arc Length

Consider a curve given by a continuous function $y = f(x)$ defined on the interval $[a, b]$, and let P be a partition of $[a, b]$.

If $y_k := f(x_k)$ and $\Delta y_k = y_k - y_{k-1}$, then the length of the curve between the points (x_{k-1}, y_{k-1}) and (x_k, y_k) is approximately

$$L_k = \sqrt{(\Delta x_k)^2 + (\Delta y_k)^2} = \sqrt{1 + \left(\frac{\Delta y_k}{\Delta x_k}\right)^2} \Delta x_k$$

when Δx_k is small. The definition of arc length is obtained by taking limit of $\sum_{k=1}^n L_k$ as $\|P\| \rightarrow 0$.



• Arc length = the limit of Riemann sums = $\sum_{k=1}^n L_k \approx \sum_{k=1}^n \sqrt{(\Delta y_k)^2 + (\Delta x_k)^2}$

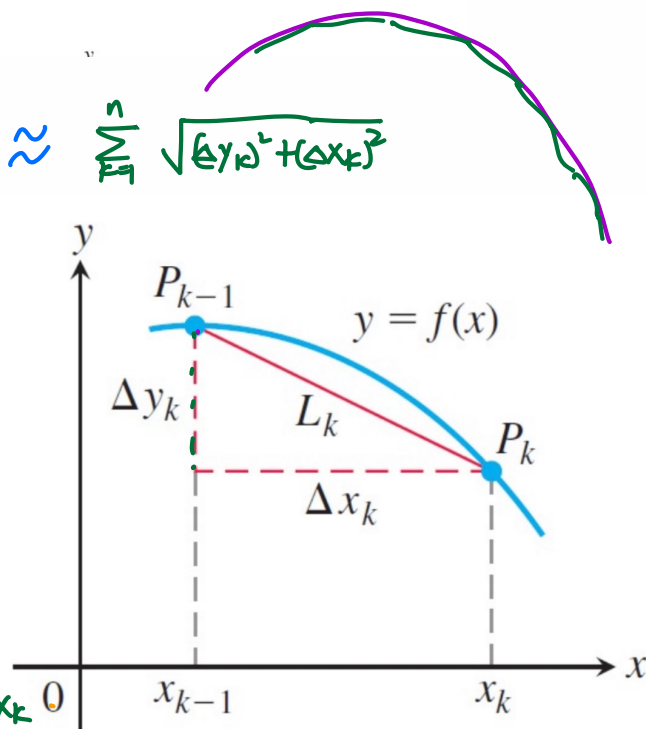
• $L_k = \sqrt{(\Delta x_k)^2 + (\Delta y_k)^2} = (\Delta x_k) \cdot \sqrt{1 + \left(\frac{\Delta y_k}{\Delta x_k}\right)^2}$

• $\frac{\Delta y_k}{\Delta x_k} = \frac{f(x_k) - f(x_{k-1})}{x_k - x_{k-1}} = f'(c_k)$

for some $c_k \in (x_{k-1}, x_k)$.

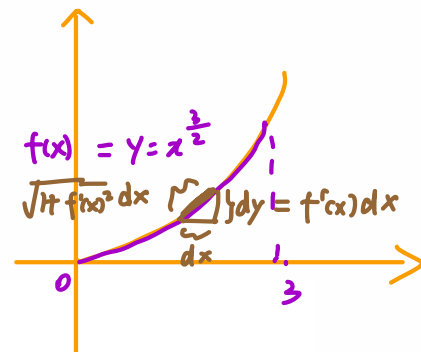
$\therefore$ Arc length = the limit of $\sum_{k=1}^n \sqrt{1 + f'(c_k)^2} \cdot \Delta x_k$

$$= \int_a^b \sqrt{1 + f'(x)^2} dx.$$



DEFINITION If f' is continuous on $[a, b]$, then the **length (arc length)** of the curve $y = f(x)$ from the point $A = (a, f(a))$ to the point $B = (b, f(b))$ is the value of the integral

$$L = \int_a^b \sqrt{1 + [f'(x)]^2} dx = \int_a^b \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx. \quad (3)$$



Example

Compute the length of the curve given by $y = x^{3/2}$, $0 \leq x \leq 3$.

Sol¹: Arc length = $\int_0^3 \sqrt{1 + f'(x)^2} \, dx$

$$= \int_0^3 \sqrt{1 + \left(\frac{3}{2}x^{1/2}\right)^2} \, dx$$

$$= \int_0^3 \sqrt{1 + \frac{9}{4}x} \, dx$$

$$= \frac{4}{9} \cdot \frac{2}{3} \left(1 + \frac{9}{4}x\right)^{3/2} \Big|_0^3$$

$$= \frac{8}{27} \left(1 + \frac{9}{4}x\right)^{3/2} \Big|_0^3$$

$$= \frac{4}{9} \cdot \frac{2}{3} 4^{3/2} \Big|_1^{1 + \frac{9}{4} \cdot 3}$$

$$= \frac{8}{27} \left(\left(\frac{31}{4}\right)^{3/2} - 1 \right) . \quad \square$$

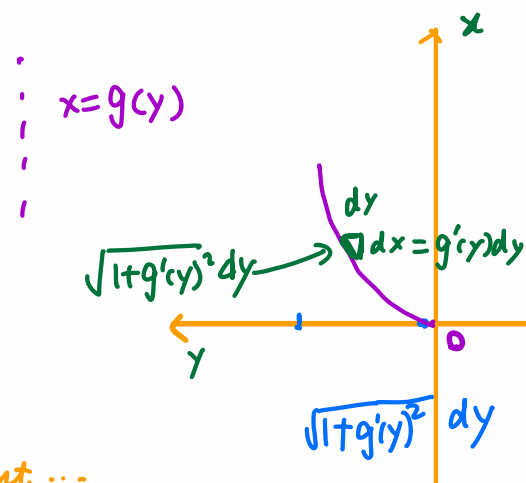
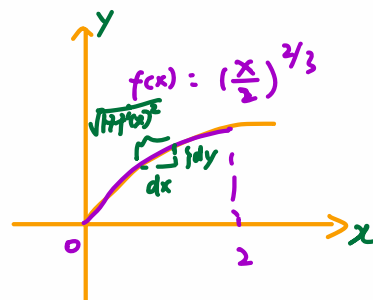
Example

Compute the length of the curve given by $y = (x/2)^{2/3}$, $0 \leq x \leq 2$.

$$\begin{aligned}
 \text{Sol}': L &= \int_0^2 \sqrt{1 + f'(x)^2} \, dx \\
 &= \int_0^2 \sqrt{1 + \left(\frac{2}{3} \left(\frac{x}{2}\right)^{-\frac{1}{3}} \cdot \frac{1}{2}\right)^2} \, dx \\
 &= \int_0^2 \sqrt{1 + \left(\frac{1}{3}\right)^2 \left(\frac{x}{2}\right)^{-\frac{2}{3}}} \, dx \\
 &= \int_0^2 \sqrt{1 + \frac{1}{9} 2^{\frac{2}{3}} x^{-\frac{2}{3}}} \, dx \\
 &= \int_0^2 \sqrt{1 + C \cdot x^{-\frac{2}{3}}} \, dx
 \end{aligned}$$

NOT easy to find its antiderivative

(There is no good way to compute it out ...)



If the curve is given by $x = g(y)$, $c \leq y \leq d$, and g' is continuous, then the arc length can be computed by

$$L = \int_c^d \sqrt{1 + \left(\frac{dx}{dy}\right)^2} \, dy = \int_c^d \sqrt{1 + (g'(y))^2} \, dy.$$

In the example above, we write

$$\begin{aligned}
 y &= \left(\frac{x}{2}\right)^{2/3}, \quad x \in [0, 2] \Rightarrow x = 2 \cdot y^{\frac{3}{2}} \triangleq g(y), \quad y \in [0, 1] \\
 \Rightarrow y^{\frac{3}{2}} &= \frac{x}{2} \Rightarrow g'(y) = \frac{3}{2} y^{\frac{1}{2}}
 \end{aligned}$$

$$\therefore L = \int_0^1 \sqrt{1 + g'(y)^2} \, dy$$

$$= \int_0^1 \sqrt{1 + (3y^{\frac{1}{3}})^2} \, dy$$

$$= \int_0^1 \sqrt{1 + 9y} \, dy$$

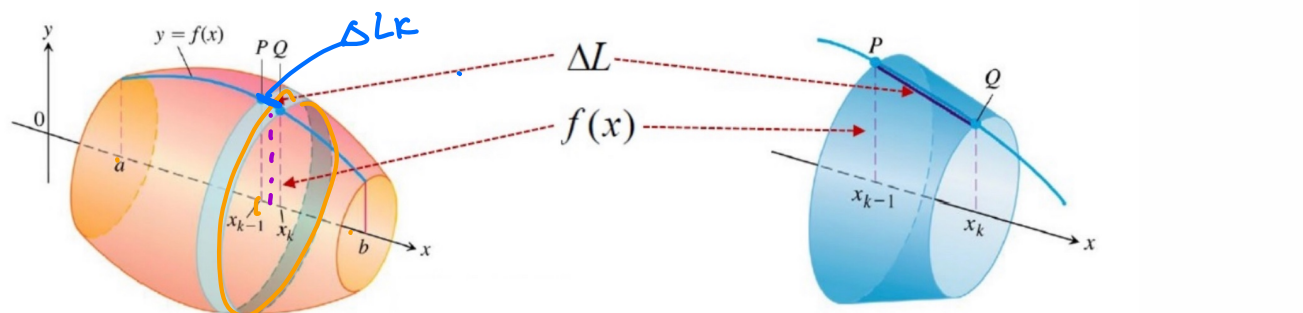
$$= \frac{1}{9} \frac{2}{3} (1+9y)^{\frac{3}{2}} \Big|_0^1$$

$$= \frac{2}{27} (10^{3/2} - 1)$$

□

Areas of Surfaces of Revolution

Consider a surface generated by revolving about the x -axis a curve $y = f(x)$, where f is ~~positive~~ *nonnegative*, for $x \in [a, b]$.



• Area of Surface of Revolution = the limit of $\sum_{k=1}^n$ k th partition of surface

= the limit of $\sum_{k=1}^n$ the length of the circle $\cdot \Delta L_k$

= the limit of $\sum_{k=1}^n 2\pi f(c_k) \cdot \Delta L_k$

= the limit of $\sum_{k=1}^n 2\pi f(c_k) \cdot \sqrt{1 + f'(c_k)^2} \cdot \Delta x_k$ where $c_k \in [x_{k-1}, x_k]$.

$$\int_a^b 2\pi f(x) \sqrt{1+f'(x)^2} dx$$

This motivates the following definition.

Definition

Let f be a ~~positive~~ ^{nonnegative} function such that f' is continuous on $[a, b]$.

The **area** S of the surface generated by revolving about the x -axis the curve $y = f(x)$, $a \leq x \leq b$, is defined by

$$S := \int_a^b 2\pi f(x) \underbrace{\sqrt{1 + \left(\frac{dy}{dx}\right)^2}}_{dL} dx = \int_a^b 2\pi f(x) \underbrace{\sqrt{1 + (f'(x))^2}}_{dL} dx.$$

$$\underline{dL} = \sqrt{1+f'(x)^2} dx$$

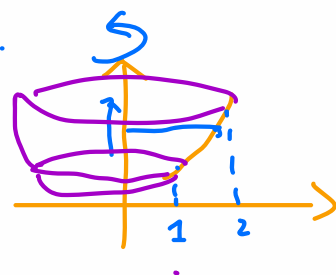
we call this "arc length differential".

Similarly, if $x=g(y)$ (g nonnegative, g' continuous on $[c, d]$) is

revolved about the y -axis for $c \leq y \leq d$, then

$$S = \int_c^d 2\pi \cdot g(y) \cdot \sqrt{1+g'(y)^2} dy.$$

e.g. The curve $y=x^2$ from $(1,1)$ to $(2,4)$ is revolved about the y -axis to obtain a surface S . Find Area(S).



Sol' : $y = x^2, 1 \leq x \leq 2 \Rightarrow x = g(y) = y^{\frac{1}{2}}, 1 \leq y \leq 4.$

$$S = \int_1^4 2\pi g(y) \sqrt{1+g'(y)^2} dy$$

$$= \int_1^4 2\pi y^{\frac{1}{2}} \sqrt{1+(\frac{1}{2}y^{-\frac{1}{2}})^2} dy$$

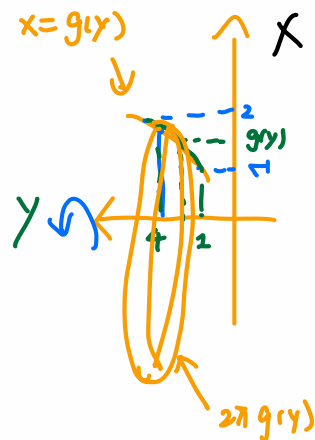
$$= \int_1^4 2\pi y^{\frac{1}{2}} (1 + \frac{1}{4}y^{-1})^{\frac{1}{2}} dy$$

$$= \int_1^4 2\pi (y + \frac{1}{4})^{\frac{1}{2}} dy$$

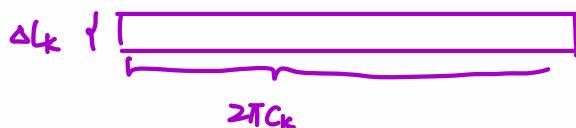
$$= 2\pi \left. \frac{2}{3} (y + \frac{1}{4})^{\frac{3}{2}} \right|_1^4$$

$$= \frac{\pi}{6} (17\sqrt{17} - 5\sqrt{5}).$$

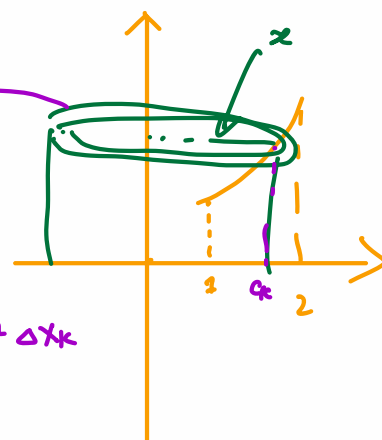
14.



Alternatively.



$$\therefore \sum_{k=1}^n 2\pi C_k \Delta L_k = \sum_{k=1}^n 2\pi C_k \sqrt{1+f'(x_k)^2} \Delta x_k$$



$$S = \int_1^2 2\pi x \sqrt{1+(2x)^2} dx$$

$$= \int_1^2 2\pi x \sqrt{1+4x^2} dx$$

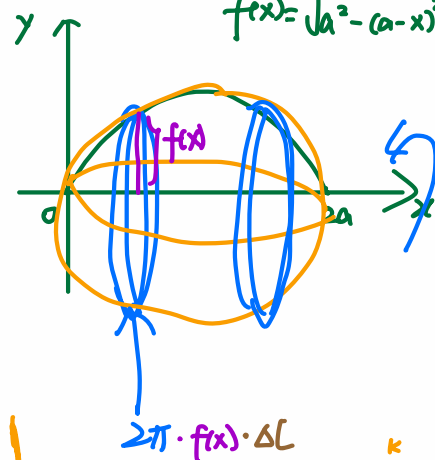
$$= \int_1^2 \pi \sqrt{1+4x^2} d(x^2)$$

$$\stackrel{y=x^2}{=} \int_1^4 \pi \sqrt{1+4y} dy = \frac{\pi}{6} (17^{\frac{3}{2}} - 5^{\frac{3}{2}}).$$

14.

E.g. Derive the surface area formula of a sphere with radius a .

$$f(x) = \sqrt{a^2 - (a-x)^2}$$



$$\text{Sol: } S = \int_0^{2a} 2\pi f(x) \cdot \sqrt{1 + f'(x)^2} \, dx$$

$$= \int_0^{2a} 2\pi \sqrt{a^2 - (x-a)^2} \left(1 + \left[\frac{1}{2} (a^2 - (x-a)^2)^{-\frac{1}{2}} \cdot (-2(x-a)) \right]^2 \right)^{\frac{1}{2}} dx$$

$$= \int_0^{2a} 2\pi \sqrt{a^2 - (x-a)^2} \left(1 + \frac{(x-a)^2}{a^2 - (x-a)^2} \right)^{\frac{1}{2}} dx$$

$$= \int_0^{2a} 2\pi \left(\sqrt{a^2 - (x-a)^2} \right) \cdot \left(\frac{a^2}{a^2 - (x-a)^2} \right)^{\frac{1}{2}} dx$$

$$= \int_0^{2a} 2\pi a \, dx$$

$$= 4a^2 \pi.$$

□.

Arc length differentials

Given $y = f(t)$ defined on some interval I , fix $a \in I$. Define

$$s(x) \triangleq \int_a^x \sqrt{1 + f'(t)^2} dt$$

for $x \in I$. Then $s(x)$ is the signed distance from $(a, f(a))$ to $(x, f(x))$ along the curve $y = f(t)$.

[When $x < a$, this signed distance is negative.]

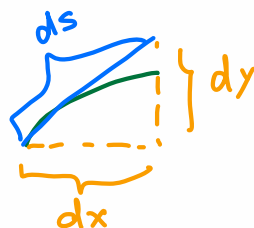
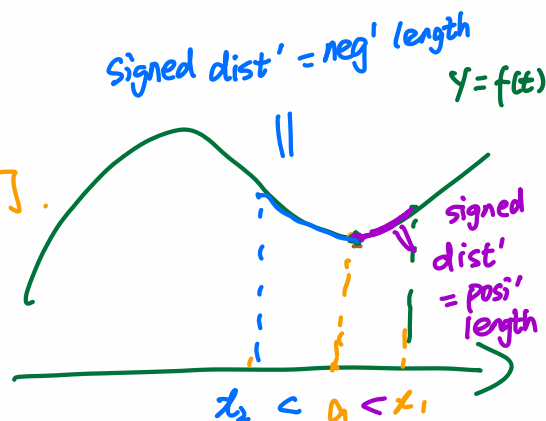
• By FTC 1, $s'(x) = \sqrt{1 + f'(x)^2} = \frac{ds}{dx}$

$\Rightarrow ds = \sqrt{1 + f'(x)^2} dx$ is called the arc length differential.

$$= \sqrt{(dx)^2 + (dy)^2}$$

$$= \sqrt{(dx)^2 + (f'(x)dx)^2}$$

$$= \sqrt{1 + f'(x)^2} \cdot dx$$



Work

- If a constant force F moves an object along a straight line for a distance d , then the work done by F to the object is $W = F \cdot d$.



DEFINITION The **work** done by a variable force $F(x)$ in moving an object along the x -axis from $x = a$ to $x = b$ is

$$W = \int_a^b F(x) dx. \quad (2)$$

Using the same idea,

the work = the limit of all Riemann sums

$$= \text{the limit of } \sum_{k=1}^n F(c_k) \cdot \Delta x_k.$$

$$= \int_a^b F(x) dx.$$

Hooke's law : the force required to stretch or compress a spring

is directly proportional to its distance x away from the natural position of the spring :

$$F(x) = k \cdot x,$$

where k is the **spring constant** with unit being force unit per length unit.

6.S.3.

EXAMPLE 3 A spring has a natural length of 1 m. A force of 24 N holds the spring stretched to a total length of 1.8 m.

- (a) Find the ~~force~~ ^{spring} constant k .
 (b) How much work will it take to stretch the spring 2 m beyond its natural length?
 (c) How far will a 45-N force stretch the spring?

Sol: (a) $F = k \cdot 0.8 = 24$

$\Rightarrow k = 30 \text{ (N/m)}$.

(b) . $W = \int_0^2 F(x) dx$

$= \int_0^2 30 \cdot x dx$

$= 15x^2 \Big|_0^2$

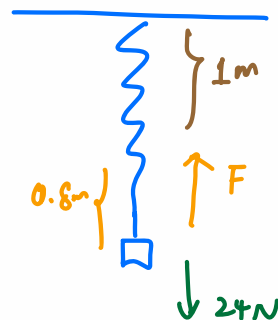
$= 60 \text{ (J)}$

(c) $F = 30 \cdot x = 45$

$\Rightarrow x = 1.5 \text{ (m)}$

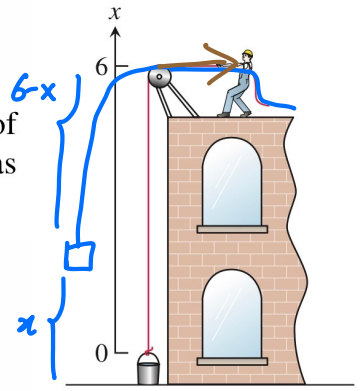
$\Rightarrow \text{Total length} = 1 + 1.5 = 2.5 \text{ (m)} .$

□ .



EXAMPLE 6.5.4

A 2-kg bucket is lifted from the ground into the air by pulling in 6 m of rope at a constant speed (Figure 6.38). The rope weighs 0.1 kg/m. How much work was spent lifting the bucket and rope?



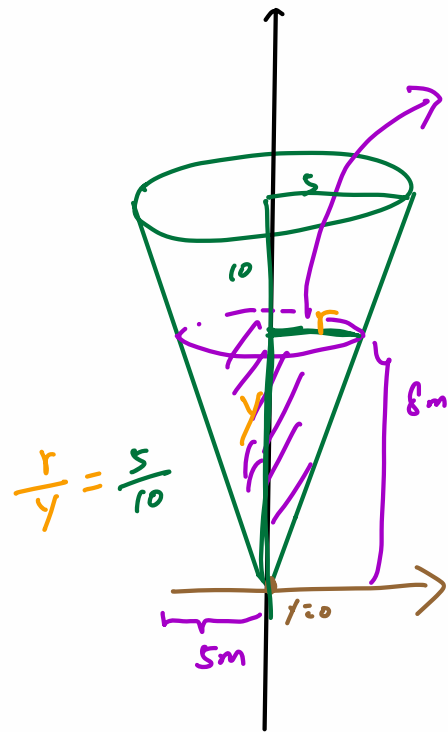
$$\begin{aligned}\text{Sol: } F(x) &= [2 + (6-x) \cdot 0.1] \cdot g \\ &= (2.6 - 0.1x) \cdot g.\end{aligned}$$

$$\begin{aligned}\therefore W &= \int_0^6 F(x) \, dx \\ &= \int_0^6 (2.6 - 0.1x)g \, dx \\ &= (2.6x - \frac{1}{2}0.1x^2) \Big|_0^6 \cdot g \\ &= 13.8 \cdot g \\ &\approx 13.8 \times 9.8 \text{ (J)}.\end{aligned}$$

□.

E.g. (Pumping liquid from container)

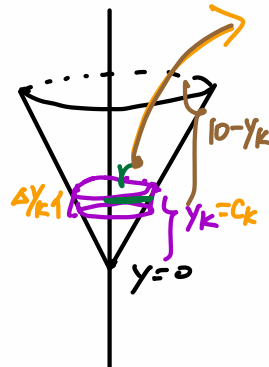
A conic tank has height 10m and base radius 5m, with vertex pointing down. The y -axis is the axis of the cone, with vertex at $y=0$. The tank is filled with olive oil with weight-density 8820 N/m^3 . The surface of the oil is 8m from the bottom of the tank. How much work is required to pump the oil to the rim of the tank?



Sol¹: ["Cut" the oil horizontally!]

The thin slice has a volume

$$\begin{aligned} \Delta y_k \cdot \pi \cdot r^2 &= \Delta y_k \cdot \pi \cdot \left(c_k \cdot \frac{5}{10} \right)^2 \\ &= \pi \left(\frac{c_k}{2} \right)^2 \Delta y_k, \text{ where } c_k \in [y_{k-1}, y_k]. \end{aligned} \quad \frac{r}{c_k} = \frac{5}{10}$$



The thin slice requires a force

$$8820 \cdot \pi \left(\frac{c_k}{2} \right)^2 \Delta y_k$$

The thin slice will be lifted for a height $10 - y_k$.

$$\therefore \text{The work is given by } \lim_{N \rightarrow \infty} \underbrace{\sum_{k=1}^N 8820 \cdot \pi \left(\frac{c_k}{2} \right)^2 \Delta y_k}_F \underbrace{(10 - y_k)}_d$$

$$\therefore \text{The work} = \int_0^8 8820 \cdot \pi \left(\frac{x}{2} \right)^2 (10 - x) dx$$

$$= 2205 \pi \int_0^8 (10x^2 - x^3) dx$$

$$= 2205 \pi \left(\frac{10}{3} 8^3 - \frac{1}{4} 8^4 \right) (\text{J}). \quad \square.$$